Github Project

ROBOT TOUR

Science Olympiad

2026

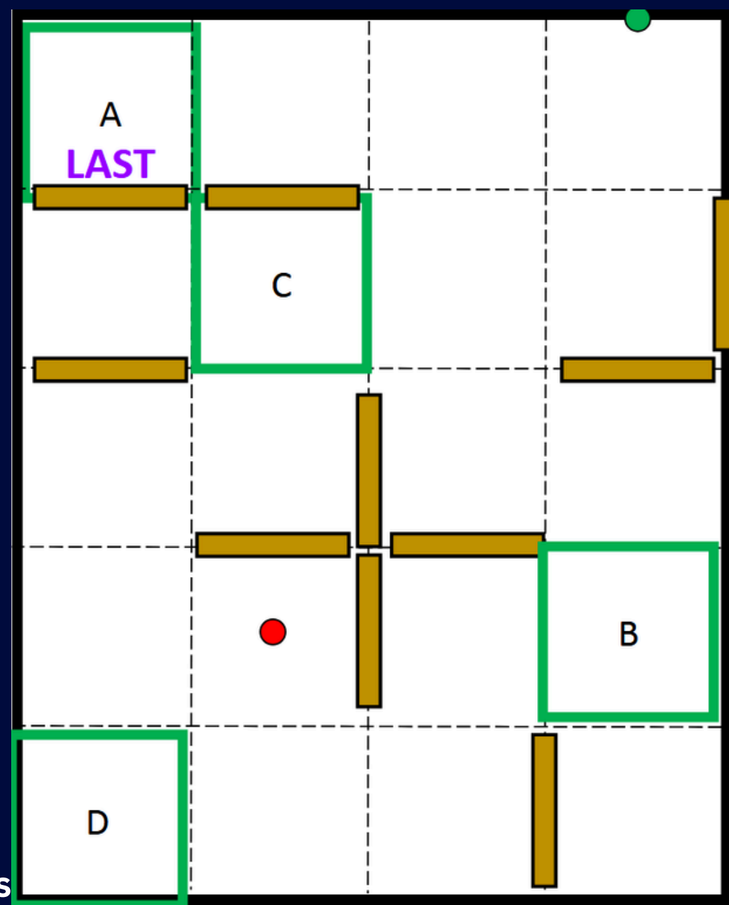
*Remember all of this Information is subject to change through each year

*made for Collinsville High School

What is it?

Robot tour is a highly technical event where a team's job is to build a robot that can move accurately and fast through a course that will be revealed to you at the time of the event.

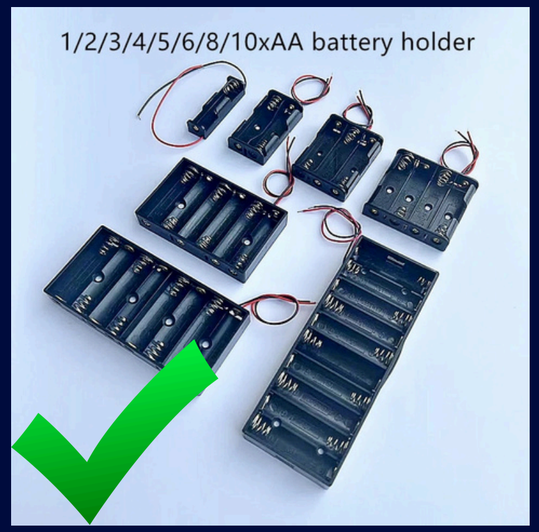
The Robot need to get from the starting point (Green Dot) to the ending point (Red Dot) in a timely and accurate fashion, without hitting the obstacles (2x4 wood rectangles). There are gates throughout the track that give you bonus points just for passing through them. In newer iterations there are weighted water bottles that give you bonus points if you move them.

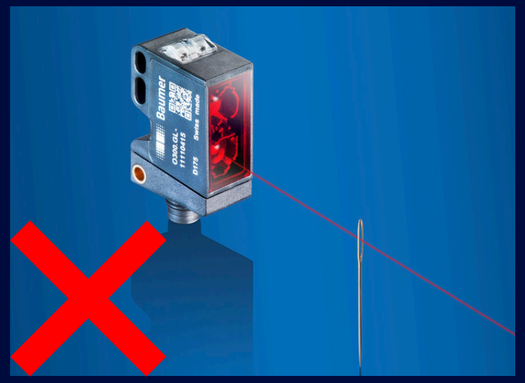


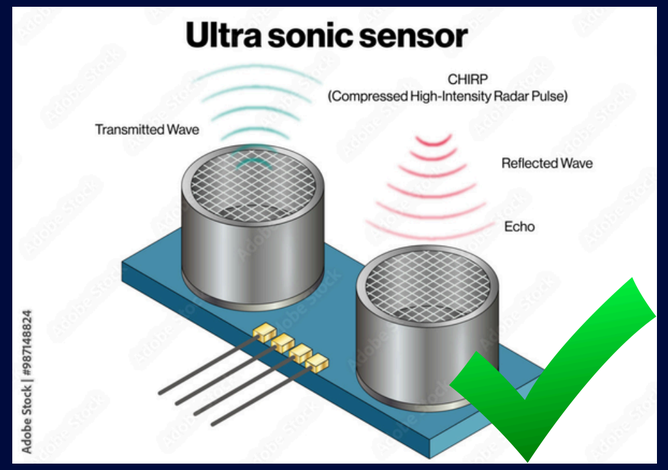
Green Dot = Start
Red Dot = End
Green Squares = Bonus Gates

Important Rules

- Robot must fit within 30cm x 30cm square by any orientation
- Robot cannot use lithium ion batteries (MUST USE AA or AAA BATTERIES (MAX 8))
- Robot cannot use any lasers sensors (alternative is ultra sonic sensors)
- Robot cannot use cameras
- Robot must have a 1/4" or 3/8" dowel rod in front as orientation mark (rod width subject to change)
- Allowed to use AI for coding the robot, but only prior to the event itself.



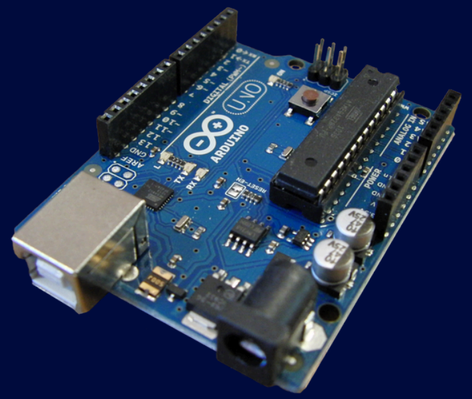




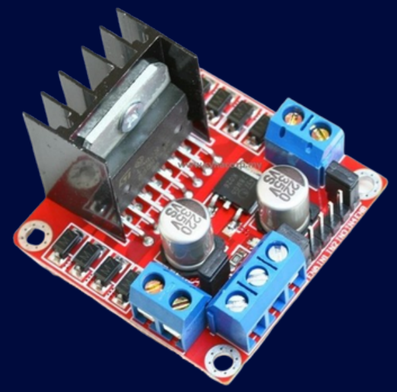
Parts I recommend

- Two motors with encoders - power the wheels (I used)
- Arduino - the brain of the robot (I used)
- Dual H Bridge Motor Controller Driver - controls motors (I used)
- Battery Source - power source (I used)
- Buck Converters - Turns higher voltages to smaller ones (I used)
- IMU - inertial measurement unit (I used)
- Chassis

*arduino

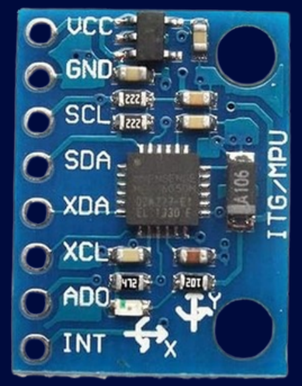


*motor driver



*buck converter

*IMU



The Motors

The motors I used were a pair of 6V 100RPM motors with 65mm wheel attachments. Both motors had optical sensors, which use light to track movement. These sensors track motor movement in units of counts/pulses/ticks. These sensors also have different resolutions, different accuracies. The units for resolution are CPR or PPR (Counts Per Revolution and Pulses per revolution)

The Function

The function of the motors is to power the movement of the robot, whether forward, backward, left, or right.

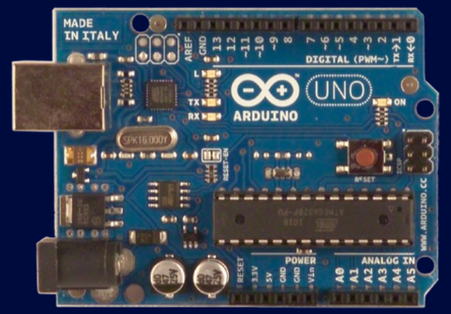
While the function of the encoders is to make sure we can ensure that the robot has an accurate movement, per whatever distance we give it (if we tell it to move 1m, it approximately moves 1m)

The Arduino

An Arduino is a popular open-source microcontroller. Open-source means that anyone can contribute to the Arduino project and code, so a platform built by a community of software engineers. A microcontroller is a microcomputer (that doesn’t have an operating system like Windows) that is used to process tasks; receive data from sensors and such; and is used to control external devices like motors. The specific Arduino used in this project is the Uno, but this project could have been downsized to the Arduino Nano

The Function

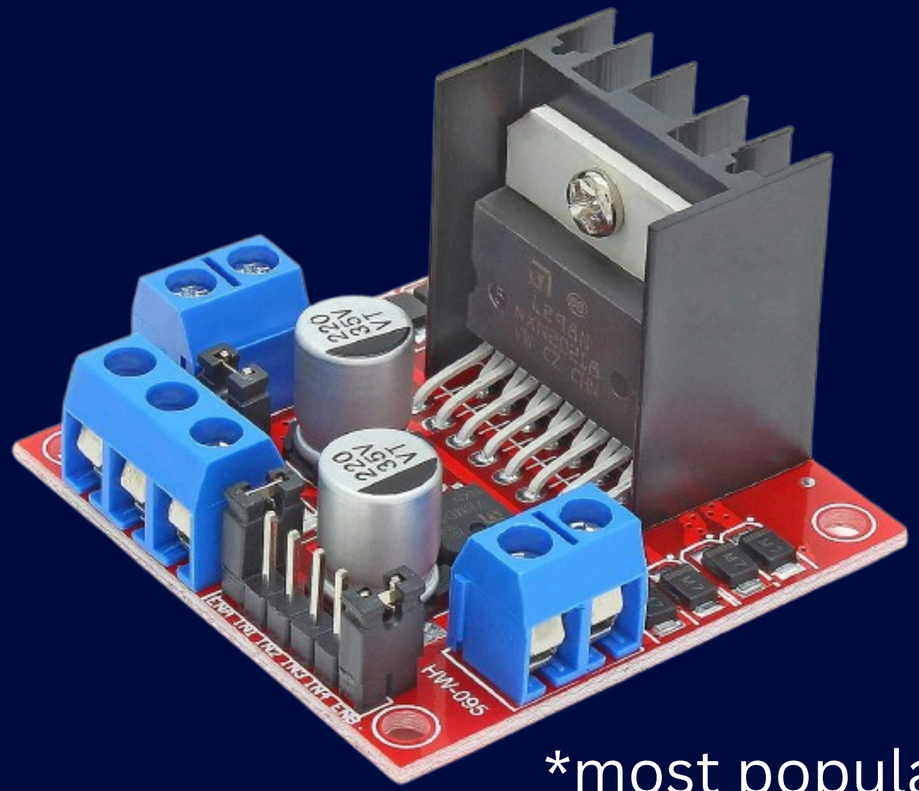
The function of the Arduino is to function as the brain of the robot, receiving the data from the motors and other sensors in this project, and is responsible for the math and computation used to control the function of the robot



The Motor Controller

The motor controller is an addition to the Arduino that features an expanded number of ports for motors and streamlines the process of controlling motors. In this situation, we will only control DC motors. This controller can also be used to move stepper motors and servo motors, but will not be used in this project.

*used in this project



*most popular version

The Power Source

The power source for this project is 8 Double A batteries connected in series so that their voltage is equal to 12volts

The Buck Converters

The buck converters are electronic boards that convert higher voltages to lower voltages. In this case, I needed one converter for the 6V motors, one converter for the 3.3V-5V IMU, and another one for the Arduino (should be 6V to 12V)

The IMU

An IMU is an electronic device used to measure and report the components' force, angular rate, and, for this project, the orientation by using the accelerometers (measures acceleration) and gyroscopes (measures angular mometum) embedded in the IMU circuit

The Function

The purpose of the IMU in this project is to track the rotation of the robot. This is used in making sure the robot is turning the full degrees set (usually in increments of 90), and making sure that when the robot moves forward, it accounts for the robot leaning off the straight path and fixes that deviation to the best of its ability.

The Chassis

The chassis is the main body of the robot that all of the components will be placed and mounted on. For my chassis, I took a slab of plexiglass and cut it to the desired robot size and drilled in holes for mounting purposes with a rotary tool

Recommended

Instead of using rotary tools and plexiglass as I did, I would recommend having the chassis 3d printed. To model the chassis, you could use an online modeling software like Tinkercad, or you could use a more GPU-based modeling software like blender or more engineering GPU-based ones like Fusion 360, AutoCAD, and SolidWorks

*for CHS students the CAVC/vocational area should have 3d printers accessible to anyone as long as you supply your own filament

SOFTWARE

Prerequisite Knowledge

- General Programming knowledge
- How to use AI for programming
- Electricity basics
- General Arduino
- Arduino Programming
- Circuits basics

Sources for Prerequisite Knowledge

- Arduino Programming Beginners Guide
- Circuit Basics

Prerequisite Hardware

- Any laptop that can download the Arduino IDE, chrome book have very selective compatibility

Design Philosophy of the Code

Variables

- Assigning motors and encoder pins
- Robot constants
- Distance Calibration Variable
- Sensor Orientation Variable

Setup Loop

Startup Libraries
Setup pins to output or inputs

Update Loop

Have the track sequence play out if a button is pressed

Main Functions

Turn Degrees
uses the IMU to record to starting angle, then moves the motors in a certain direction (for 90, right goes forward and left goes back) to turn in the wanted direction until the IMU reads the desired angle.

Distance Function
Applies power to motors and tracks the ticks of the encoders. Once the encoders reach the estimate for whatever the inputted distance would be, the robot stops. Also automatically straightens robot to account for drift

Libraries

MPU6050_tockn.h - library used for communication with IMU
Wire.h - library used to communicate with I2C

*Code is uploaded on Github